A Shockingly Simple Proof of the Riesz Representation Theorem on Finite-Dimensional Spaces

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1 Abstract

The Riesz Representation Theorem is an important result in linear algebra that allows for the definition of the adjoint. The proof exhibited in this short paper is heavily inspired by Exercise 1-12 in *Calculus on Manifolds* by Michael Spivak [Spi65]. No real mathematical insight has come from the author of this paper, apart from the (very trivial) generalisation of the proof from $\mathbb{R}^n$ to an abstract finite-dimensional inner product space V.

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2 Proof

Riesz Representation Theorem. Let V be a finite-dimensional inner product space, and φ a linear functional on V. Then there exists a unique vector $v \in V$ such that $\varphi(u) = \langle u, v\rangle$, for all $u \in V$.

Proof: For each $v \in V$, define $\psi_v \in V'$ by $\psi_v(u) = \langle u, v\rangle$, for all $u \in V$. Define the linear map $T : V \to V'$ by $Tv = \psi_v$ for all $v \in V$. We claim that T is a bijection from V onto its dual space. Indeed, suppose that $v \in V$ is such that $Tv = 0$. Then in particular, we have the relation

$$\psi_v(u) = \langle u, v\rangle = 0 \tag{1}$$

for all $u \in V$. Choosing $u = v$ in Equation 1, we obtain

$$\psi_v(v) = \|v\|^2 = 0$$

so that $v = 0$ and T is injective as $\ker T = \{0\}$

Since $\dim V = \dim V'$, we have that T is surjective by the Rank-Nullity Theorem. Let $\varphi \in V'$ be given. Our discussion above shows that there exists a unique vector $v \in V$ such that

$$\langle u, v \rangle = [Tv](u) = \varphi(u), \tag{2}$$

for all $u \in V$, which is the statement of the Theorem. ■

3 Note: Relation to the Cross Product

We can use the Riesz Representation Theorem to define the cross product, as follows. Consider $\mathbb{R}^n$, and suppose the vectors $v_1, \dots v_{n-1} \in \mathbb{R}^n$ are given. Define the map $\varphi : V \to \mathbb{R}$ by

$$\varphi(w) = \det\left(v_1, \dots v_{n-1}, w\right),$$

where of course the matrix is taken with respect to the standard Euclidean basis of $\mathbb{R}^n$.

We note that since $\det \in \Omega^n(\mathbb{R}^n)$, we have that $\varphi \in V' = \Omega^1(V)$. Invoking the Riesz Representation Theorem, we see that there exists a unique vector $u \in V$ such that, for all $w \in V$,

$$\varphi(u) = \langle w, u \rangle = \det\left(v_1, \dots, v_{n-1}, w\right). \tag{3}$$

We say that u is the *cross product* of $v_1, \dots, v_{n-1}$, and we write this as $u = v_1 \times \cdots \times v_{n-1}$.

Since the determinant is an alternating multilinear form, we see that if σ is a permutation of the integers from 1 to n,

$$v_{\sigma(1)} \times \cdots \times v_{\sigma(n-1)} = \operatorname{sign}(\sigma)(v_1 \times \cdots \times v_{n-1}).$$

In $\mathbb{R}^3$, this leads to the familiar formula from highschool that $\vec{a} \times \vec{b} = -\vec{b} \times \vec{a}$. The multilinearity of the determinant allows us to deduce more nice properties about the cross product, such as

$$v_1 \times \cdots \times (av_j) \times \cdots \times v_{n-1} = a(v_1 \times \cdots \times v_{n-1}).$$

References

[Spi65] Michael Spivak. *Calculus on Manifolds*. Mathematics Monograph Series. CRC Press, 1965. ISBN: 9780805390216.